

Smart Crop Care & Farmer Advisory: An Integrated AI-Based Farm Decision Support System with Web and Flutter/Dart Mobile Interfaces

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Abstract—Abstract— Crop selection is more than nutrients in the soil or weather alone. They also need to consider the need of fertiliser, crop symptoms, area to be cultivated and cost of cultivation.

In this paper we introduce the Smart Crop Care Farmer System, a farm decision support system developed using a FastAPI backend, a web interface and Flutter/Dart mobile client support. The model uses a Random Forest for classify crops, an Extra Trees classifier for fertilizers for fertilizers, and an estimation of the cost using a Random with a deterministic formula. The disease support module is designed to use a ResNet50 transfer-learning model when trained disease artifacts are found; otherwise, it reverts to clear advisory fallback. Manual values are entered into the levels of nitrogen, phosphorus, potassium, temperature, pH, rainfall, moisture, soil type, land area and optional crop leaf image. It provides crop prediction, alternative recommended alternative crops, fertilizer recommendation, field readiness, disease advisory, operational notes and downloadable reports. This work is a project implementation implementation that links machine learning models withfarmer-friendly web and mobile apps. **Index Terms** - Precision Agriculture, Crop Recommendation, Random Forest, Extra Trees, Cost Estimation, Disease Detection, ResNet50, FastAPI, Flutter, Dart, Fertilizer Recommendation.

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I. INTRODUCTION

Agricultural decision-making involves agronomic, environmental and economic data. The farmer needs to decide the crop, fertilizer, detect disease early and decide the cost of cultivation. Many times, farmers make these decisions based on personal experience, fragmented data or different sources of advice. This may lead to an incorrect crop choice, unbalanced fertiliser use and poor cost management.

India has a large number of farmers and many small farmers may not have access to expert advice. Web technologies,

using machine learning systems, can assist by transforming farm details into more readable and comparable information. An effective advisory tool should not be limited to crop forecasting. It should link the crop selection to fertiliser advice, disease management, cost planning and record keeping.

In this paper, we present **Smart Crop Care & Farmer Advisory**, an AI-powered farm management tool. The system uses a FastAPI backend, a static web dashboard with HTML, CSS and JavaScript, and a Flutter/Dart mobile app with backend API calls. It uses Scikit-learn pipelines for crop recommendation, fertilizer recommendation, and cost estimation. It also supports TensorFlow ResNet50 for disease detection (optional). Farm inputs are manually entered through the web or mobile app. Smart presets are available for trials and demonstrations.

The main contributions of this work are:

- A comprehensive farm advisor dashboard with inputs of soil, weather, moisture, soil type, crop image, and area.
- A **Random Forest crop recommendation model** with nutrient and environmental features and further feature engineering.
- An **Extra Trees fertilizer recommendation model** using crop, moisture, soil, and nutrients.
- A **Random Forest-based cost estimator pipeline** for total cost, revenue, profit, cost per acre, profit margin (with a deterministic fallback formula).
- A **ResNet50-based disease support system** that uses disease artifacts to predict leaf images.
- A FastAPI API server with user login, report saver, report download, presets, soil type chips, pH tone, image preview, report vault statistics, theme changer, and Flutter/Dart mobile API.

The rest of this paper is structured as follows: Section II reviews related literature. Section III provides an overview of the system architecture. Section IV describes the proposed workflow. Section V gives the mathematical formulation. Section VI presents implementation and results. Section VII discusses social and practical implications. Section VIII concludes the paper and discusses future work.

II. LITERATURE SURVEY

There have been a number of applications of machine learning in agriculture such as soil nutrient analysis, crop recommendation, fertilizer recommendation, disease diagnosis and web-based advisory systems. The studies below assisted in the identification of the required inputs and modules for the system.

A. Soil Nutrient and Sensor Systems

Cheruvu et al. [1] proposed a smart farming system with NPK sensors and demonstrated that nutrient data can be used in crop and fertilizer recommendations. Agarwal et al. [3] investigated soil-based crop recommendation and the importance of pH and NPK values. Abhinov et al. [10] developed a machine learning-based soil-based crop recommendation system. These papers demonstrate that soil nutrients and pH are valuable data in a crop recommendation system.

B. Environment in Crop Prediction

Shunmuga Lakshmi et al. [2] proposed a crop prediction system given different weather conditions. Their research demonstrates that crops are suitable for particular environmental and soil values. As such, the proposed system uses manual inputs of temperature, humidity and rainfall.

C. Ensemble Learning for Agricultural Classification

Random Forest and other ensemble techniques can be applied in agricultural classification as they can capture the complexity and nonlinearity of feature relationships. Deshmukh et al. [4] discussed crop selection and prediction using machine learning methods. M. B. et al. [5] also explored crop analysis with soil and environmental attributes. These studies justify the recommendation of Random Forest for crop prediction.

D. Fertilizer Recommendation and Cost Planning

Ali et al. [8] suggested crop and fertilizer recommendations using machine learning. Basavaraju et al. [11] presented yield prediction and fertilizer usage in smart agriculture. Vaddi et al. [13] discussed crop and fertilizer recommendations with yield prediction. These studies suggest that crop recommendation is more valuable when accompanied with fertilizer and economic advice.

E. Leaf Disease Detection and Web Applications

Harish and Devi [6] talked about the integration of IoT and machine learning in precision agriculture for crop recommendation and leaf disease identification. Kumar et al. [12] discussed an AI-based plant disease recognition and crop recommendation system. Kethineni et al. [14] strongly recommended web-based

crop, fertilizer and pesticide recommendation. These works justify the use of disease image processing and web interface. An additional Flutter/Dart client for mobile is integrated in this project.

F. Gap Analysis and Motivation

It can be observed from the literature that many systems concentrate on performing one main function. A few crop, others fertilizers, some disease. The proposed system integrates these features into one application that recommends crops, fertilizers, provides cost estimates, supports disease diagnosis, generates reports, and includes a user login system.

Fig. 1 illustrates the architecture of the system. It can be broken down into three sections: the Data Input Section, the Machine Learning Section and the Deployment Section. The input side takes user field values as input. The middle part does the preprocessing, feature engineering and prediction. The output side displays the advisory in the web dashboard and Flutter/Dart mobile app.

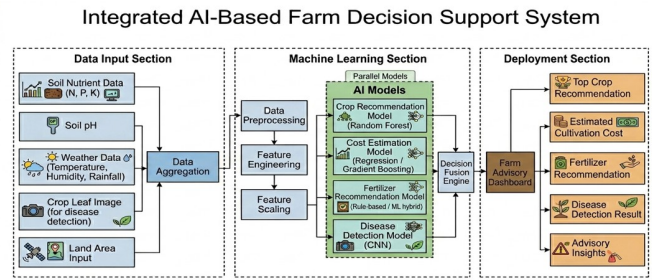


Fig. 1. Architecture of Integrated AI-Based Farm Decision Support System.

G. Data Input Section

This Section is used to input the required values to the advice workflow. The implemented system accepts:

- **Nutrient Concentrations:** Nitrogen, phosphorus and potassium.
- **Soil pH:** pH of the field.
- **Weather Data:** Temperature, humidity and rainfall values that the farmer inputs through the portal.
- **Crop Leaf Image (Optional):** Uploaded image with diseases.
- **Area Cultivated:** Area in acres for the cost benefit analysis.
- **Soil Moisture and Type:** Additional factors for fertilizer recommendations.

Weather values are input manually or demonstration presets are used. There are also soil type chips to select, a pH slider with labels such as acidic, neutral or alkaline, and smart presets that populate values of typical farms.

H. Machine Learning Section

Machine Learning Section includes data preprocessing, feature engineering and inference. Numerical features are imputed

and scaled if necessary. Categorical features like soil type and crop type are encoded. The implemented models are:

- **Crop Recommendation Model:** Random Forest classifier.
- **Fertilizer Recommendation Model:** Extra Trees classifier.
- **Cost Calculator Model:** Random Forest regressor, with seed-formula backup.
- **Disease Model:** Optional ResNet50 transfer-learning CNN.

I. Deployment Section

The Deployment Section includes the web dashboard and the Flutter/Dart client. The web dashboard shows the main crop recommendation, other crop cards, fertilizer recommendation, fertilizer plan, cultivation cost, rate of return (ROI), margin, readiness score, recommendation signal, disease support result, disease action plan, crop advisory insights, reports, and storage details. The Dart client makes HTTP requests to the same FastAPI endpoints for crop recommendation, fertilizer recommendation, price estimation and disease support. This means that the same code can be called from the web and mobile clients.

III. PROPOSED METHODOLOGY

We have a pipelined service for an advisory system. Every phase receives a specified input and has a specified output. This allows us to connect the values of the farm to the dashboard.

Phase 0: Data Collection and Preparation

The user enters the values for NPK, temperature, humidity, pH, rainfall, moisture, soil type and area in the dashboard or mobile app. The server converts the numbers to model-specific data objects. The crop pipelines imputes the median for missing entries and performs feature engineering. The cost and fertilizer pipelines do their preprocessing for numerical and categorical features.

A. Phase 1: Crop Classification with Random Forest

The crop classifier is given soil and environmental features to recommend a crop. Random Forest is selected because it takes into account complex interactions between nutrients, pH, temperature, humidity, and rainfall without enforcing a strict rule.

B. Phase 2: Fertilizer Recommendation Using Extra Trees

The fertiliser module is given the estimated crop and other information such as temperature, humidity, moisture, soil type and nutrients. Extra Trees is employed to support both continuous and discrete features following the feature engineering.

C. Phase 3: Support for Disease Using ResNet50 CNN

The disease module accepts an optional leaf image. If the ResNet50 trained models and labels exist, the image is classified and associated with a message. If the image or model is not available, the default message is returned, but the diagnosis is not forced.

D. Phase 4: Economic Module for Economic Advice

This module gives the economic analysis for the crop and size of the field. This includes the total cost, total revenue, total profit, cost per acre and profit margin. The backend uses the deterministic cost functions in Section V if the regressor is not trained.

E. Phase 5: Recommendation integration and report

The dashboard displays the results of the crop, fertilizer, cost and disease modules. The user interface displays these as different crop combinations, fertilizer timings, planting time, signal, action notes, disease cards and risk cards. The Report module stores the input and output as a JSON and generates an HTML report.

F. Step 6: Flutter/Dart Client

The mobile application is developed using Flutter/Dart. It has a Dart service layer that communicates with the FastAPI backend via HTTP POST requests with JSON payload. Here, the mobile app adopts the same crop advise, fertiliser advise, price estimate and disease advise models developed in the backend rather than attempting to re-develop the machine learning models.

Algorithm 1 Integrated Advisory Generation

```

1: Input from farm  $N P K T H pH R$  moisture soil type acres
   (optional leaf image)
Ensure: Recommended crop, recommended fertilizer, finan-
   cial forecast, recommended disease solution and report
2:  $X_c \leftarrow [N, P, K, T, H, pH, R]$ 
3:  $\hat{C} \leftarrow \text{RandomForestCropModel}(X_c)$ 
4:  $X_f \leftarrow [T, H, \text{moisture}, N, P, K, \text{soil}, \hat{C}]$ 
5:  $\hat{F} \leftarrow \text{ExtraTreesFertilizerModel}(X_f)$ 
6:  $\text{Finance} \leftarrow \text{CostEstimator}(\hat{C}, \text{soil}, \text{acres})$ 
7: if leaf image and disease model is available then
8:    $\hat{D} \leftarrow \text{ResNet50DiseaseModel}(\text{image})$ 
9: else
10:   $\hat{D} \leftarrow$  structured fallback disease advisory
11: end if
12:  $\text{Report} \leftarrow$  log JSON/HTML input and output
13:  $\text{MobileClient} \leftarrow$  call advisory service using Flutter/Dart
   service layer
14: return  $\{\hat{C}, \hat{F}, \text{Finance}, \hat{D}, \text{Report}\}$ 

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IV. MATHEMATICAL FORMULATION

Here we present the main formulas of the processes. The equations are from the code's feature engineering and fallback functions.

A. Crop Feature Engineering

The feature vector of the crop recommendation model is the original:

$$X_c = [N, P, K, T, H, pH, R] \quad (1)$$

where X_c is the vector of crop model inputs. The NPK are the three main nutrients in the soil: nitrogen (N), phosphorus (P) and potassium (K). The variables T , H and pH and R

are temperature, humidity, soil acid or alkalinity and rainfall, respectively.

The engineered features for nutrients and interactions are:

$$NPK_{total} = N + P + K \quad (2)$$

NPK_{total} is the total of the macronutrients (nitrogen, phosphorus and potassium).

$$\text{Ratio}_{NP} = \frac{N}{P + \epsilon}, \text{Ratio}_{NK} = \frac{N}{K + \epsilon}, \text{Ratio}_{PK} = \frac{P}{K + \epsilon} \quad (3)$$

The terms Ratio_{NP} , Ratio_{NK} and Ratio_{PK} are the ratios of nitrogen to phosphorus, nitrogen to potassium and phosphorus to potassium, respectively. The variable ϵ is a small number to avoid division by zero.

$$\text{THI} = T \times \frac{H}{100} \quad (4)$$

where THI is the temperature-humidity interaction. This standardises the humidity and multiplies this by the temperature.

$$\text{RPI} = R \times pH \quad (5)$$

where RPI is the rainfall and pH interaction which is the rainfall and acidity effect on crop suitability.

$$\text{MBI} = H + \frac{R}{10} \quad (6)$$

where MBI is the moisture balance index in the crop feature engineering step. The above equations are in line with the crop feature engineering.

B. Random Forest Crop Classification

Let M be the number of trees in the Random Forest. Crop class selection is made using majority vote:

$$\hat{C} = \arg \max_{c \in C} \sum_{m=1}^M \mathbf{1}(T_m(X_c) = c) \quad (7)$$

In the equation above, $\hat{C}$ is the predicted crop, C is the set of all crop classes, T_m is the m -th decision tree, M is the total number of decision trees and $\mathbf{1}(\cdot)$ is the indicator function, which is 1 if the condition is true and 0 if it is false.

The class confidence can be defined as:

$$P(\hat{C}|X_c) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}(T_m(X_c) = \hat{C}) \quad (8)$$

And $P(\hat{C}|X_c)$ is the confidence for the predicted crop. This is the number of trees that voted for C .

C. Fertilizer Feature Engineering

The fertilizer model is fed continuous and discrete features:

$$X_f = [T, H, M_s, N, P, K, S_t, \hat{C}] \quad (9)$$

The inputs to the fertilizer model are X_f , M_s is soil moisture, S_t is soil type and $\hat{C}$ is the predicted crop from the crop model.

The numeric engineered features are:

$$\text{Nutrient}_{total} = N + P + K \quad (10)$$

where Nutrient_{total} represents the value of the nutrients in the fertilizer.

$$\text{Ratio}_{NP}^f = \frac{N}{P + \epsilon}, \text{Ratio}_{PK}^f = \frac{P}{K + \epsilon} \quad (11)$$

and Ratio_{NP}^f and Ratio_{PK}^f are nutrient balance ratios for a given fertilizer.

$$\text{MTI} = M_s \times T, \quad \text{HTI} = H \times T \quad (12)$$

where MTI is the moisture-temperature interaction and HTI humidity-temperature interaction.

$$\text{Density}_{nutrient} = \frac{\text{Nutrient}_{total}}{M_s + \epsilon} \quad (13)$$

where $\text{Density}_{nutrient}$ is the density of nutrients in the soil moisture. The moisture is added to 1 to avoid division by zero.

The soil-crop interaction is a categorical variable given by:

$$\text{SCP} = \text{concat}(S_t, \hat{C}) \quad (14)$$

where SCP is the soil-crop pair feature and is the combination of soil type and predicted crop type.

After the pre-processing step, the Extra Trees algorithm predicts:

$$\hat{F} = \text{ExtraTrees}(X_f) \quad (15)$$

and $\hat{F}$ is the fertilizer recommendation.

D. ResNet50 Disease Classification

To assist with disease, the image is resized to 224×224 pixels. If there are trained artifacts, ResNet50 generates a feature vector z . The final classifier gives class probabilities:

$$\mathbf{p} = \text{softmax}(Wz + b) \quad (16)$$

Here, $\mathbf{p}$ is the vector of probabilities of disease classes, z is the feature vector of the ResNet50 model, W corresponds to the last layer weights and b the bias.

The predicted disease label is:

$$\hat{D} = \arg \max_{d \in \mathcal{D}} p_d \quad (17)$$

where D is the predicted disease label, $\mathcal{D}$ is the set of disease labels, and p_d is the probability of disease label d .

E. Estimating Cost and Revenue

Cost prediction uses direct cost features and other derived features. The cost of input per acre is:

$$C_{acre} = C_{seed} + C_{fert} + C_{labor} + C_{machinery} + C_{water} \quad (18)$$

with C_{acre} the total cost of inputs per acre; C_{seed} , C_{fert} , C_{labor} , $C_{machinery}$ and C_{water} the seed, fertilizer, labor, machinery and water costs per acre.

The additional financial features are:

$$A_{log} = \log(1 + \max(A, 0.1)) \quad (19)$$

Here A is the size of the land in acres and A_{log} is a log transformed size of land feature.

$$V_{yield} = Y_{acre} \times P_{market} \quad (20)$$

where V_{yield} is the value yield per acre, Y_{acre} is the yield per acre and P_{market} is the market price per kilogram.

$$Share_{water} = \frac{C_{water}}{C_{acre} + 1} \quad (21)$$

where $Share_{water}$ is the share of water cost in the total cost of inputs. The 1 in the denominator is to avoid a division by 0.

$$Ratio_{mech} = \frac{C_{machinery}}{C_{labor} + 1} \quad (22)$$

and $Ratio_{mech}$ is the ratio of machinery cost to labor cost.

When the fallback formula is used, total cost, revenue and profit are calculated as:

$$C_{total} = A \times C_{acre} \quad (23)$$

where C_{total} is the total cost to grow the area selected.

$$R_{expected} = A \times Y_{acre} \times P_{market} \quad (24)$$

where $R_{expected}$ is the expected revenue from the chosen area.

$$P_{expected} = R_{expected} - C_{total} \quad (25)$$

where $P_{expected}$ is the expected profit (expected revenue - total cost).

$$Margin = \frac{P_{expected}}{R_{expected}} \quad (26)$$

with $Margin$ the profit margin. For implementation safety the margin is set to zero if expected revenue is zero.

The dashboard also shows return on investment:

$$ROI = \frac{P_{expected}}{C_{total}} \quad (27)$$

with ROI being the profit-to-cost ratio. If cost is zero, the value of the ROI is zero.

F. Dashboard Readiness Score

The frontend calculates a readiness score to give an idea of whether the farm profile is promising. It takes into account pH, temperature, humidity, rainfall, moisture, nutrient balance and profitability:

$$Readiness = 100 \times (0.20S_{pH} + 0.14S_T + 0.12S_H + 0.12S_R + 0.14S_M + 0.12S_N + 0.16S_P) \quad (28)$$

and S_{pH} is the pH comfort score, S_T is the temperature score, S_H is the humidity score, S_R is the rainfall score, S_M is the moisture score, S_N is the nutrient balance score, and S_P is the profit outlook score. The readiness is shown as a percentage in the dashboard. In the implementation, the component scores are capped before displaying the readiness percentage so that extreme input values do not result in unrealistic readiness percentages.

V. IMPLEMENTATION AND RESULTS

A. Experiments and Technology Stack

The server is written in Python and is based on FastAPI. The frontend web application is written in HTML, CSS and JavaScript and is served as static files. The mobile app is implemented in Flutter using Dart with an API service layer which communicates with FastAPI via HTTP requests. Scikit-learn is used for the crop, fertilizer and cost estimation models. We use joblib for storing model artifacts. TensorFlow is used for the ResNet50 disease detection, if enabled. Users, sessions and reports are stored as JSON files.

The user interface gives temperature, humidity, rainfall, soil, soil moisture, NPK levels and the area of land. Currently the weather data are entered or preset.

TABLE I
IMPLEMENTED TECHNOLOGY STACK

Component	Implementation
Backend	Python, FastAPI
Web frontend	HTML, CSS, JavaScript
Mobile frontend	Flutter, Dart
Mobile service client	Dart HTTP service layer
Crop model	Random Forest Classifier
Fertilizer model	Extra Trees Classifier
Cost model	Random Forest Regressor / formula fallback
Disease module	ResNet50 (transfer learning) CNN support
Model storage	Joblib, Keras file
Report storage	JSON and generated HTML
Authentication	PBKDF2-HMAC based password hashing, bearer sessions

B. Model Scope Verification

This paper describes only the model features that are implemented in the project:

- The crop recommendation model is a Random Forest classifier.
- The fertilizer recommendation model is correctly described as an Extra Trees classifier.

- The financial forecast model is correctly described as a Random Forest regressor with a deterministic formula.
- The disease module is correctly described as a ResNet50 transfer-learning CNN pipeline, with artifacts trained.
- Ten-year weather intelligence, React deployment and multi-language advisory are not claimed as features.

C. API Endpoints

TABLE II
KEY ENDPOINTS HIT FROM WEB AND FLUTTER/DART CLIENTS

Endpoint	Purpose
/auth/login	User login
/auth/register	User registration
/auth/me	User profile
/predict/crop	Crop recommendation
/predict/fertilizer	Fertilizer recommendation
/estimate/price	Financial projection
/detect/disease	Disease support
/workspace/advisory	Composite advisory workflow
/workspace/disease	Disease workspace workflow
/reports/save	Save advisory report
/reports	List reports
/reports/{report_id}/download	Download as HTML report

D. Dashboard Screenshots

The following screenshots can be included in the figures folder of the paper.

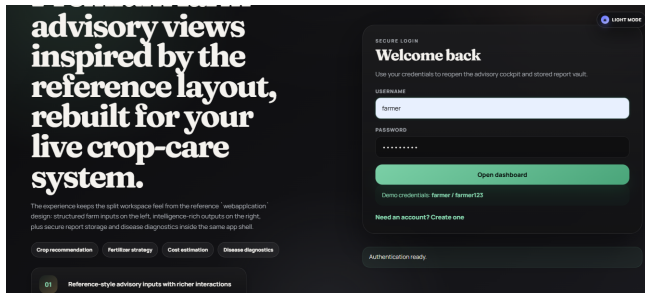


Fig. 2. Login/registration page.

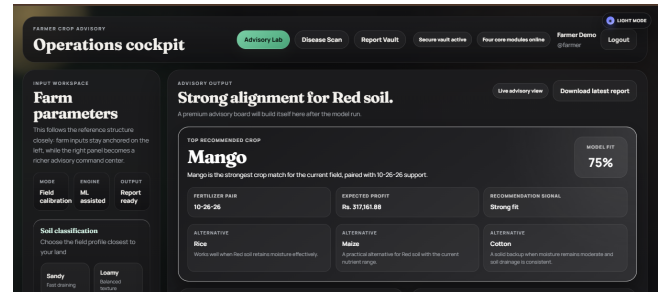


Fig. 4. Crop recommendations and fertilizer support.

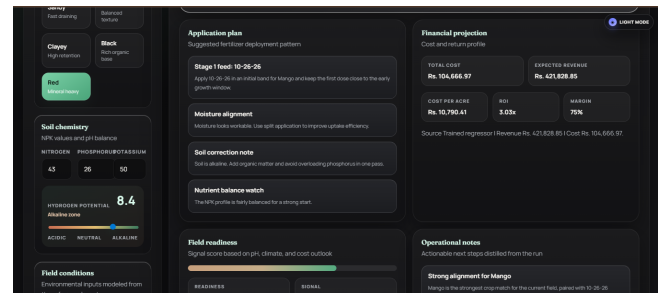


Fig. 5. Financial projection output.

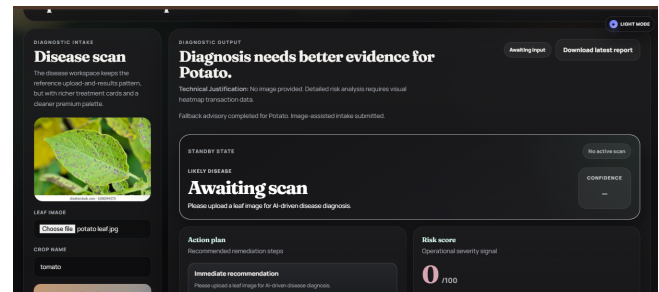


Fig. 6. Disease support workspace.

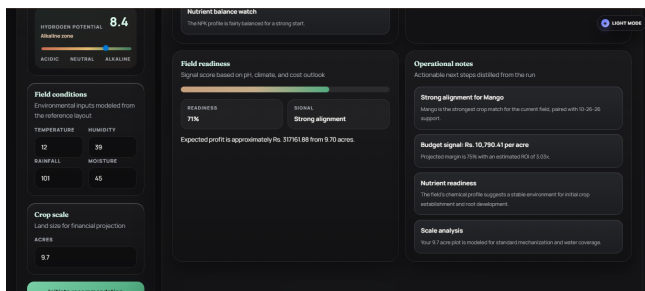


Fig. 3. Farm advisory input form.

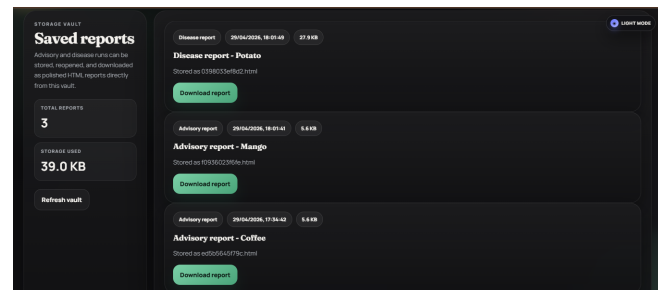


Fig. 7. Report vault and downloaded report support.

E. System Output

The system has the following main output:

- 1) best crop for planting,
- 2) suggested fertilizer and plan of application,
- 3) financial plan with return and margin,
- 4) alternative crop recommendations and notes,
- 5) readiness score and recommendation for field,
- 6) disease support with risk and action plan,
- 7) a downloadable or stored advisory (disease) report.

These are output on the cards in the dashboard, and can also be accessed by the Flutter/Dart mobile app via the API. The financial projection card displays total cost, revenue, profit, cost per acre, return on investment (ROI) and profit margin. The disease workspace includes image preview, technical note, confidence (if applicable), risk score and action plan. The report vault is used to store advices and diagnostics, it shows the total number of reports and storage used and downloads the latest report.

VI. SOCIAL AND ECONOMIC IMPACT

A. Farmer Decision Support

The system helps users read soil and environmental values in one place instead of comparing them manually across different sources. The dashboard presents crop recommendation, fertilizer recommendation, cost estimation, disease support, and advisory summary together.

B. Financial Awareness

The cost estimation module gives an approximate view of cultivation cost, expected revenue, and profit. This can support early financial planning before investing in seeds, fertilizer, labor, machinery, and water.

C. Disease Awareness

The disease support workflow allows users to upload leaf images and receive a disease-related response when the disease model is available. If the model or valid image is unavailable, the fallback message clearly explains that a trained model or suitable image is required.

D. Report-Based Traceability

Saved HTML reports make the system useful for project demonstration and advisory record keeping. A farmer, student, or agricultural officer can download and review previous recommendations. The Flutter/Dart client improves access because the same advisory services can be reached from a mobile interface.

VII. CONCLUSION AND FUTURE WORK

This paper presented the Smart Crop Care & Farmer Advisory System, an AI-based farm decision support application with web and Flutter/Dart mobile access. The implemented system combines Random Forest based crop recommendation, Extra Trees based fertilizer recommendation, Random Forest based cost estimation with formula fallback, and optional ResNet50-based disease support. The application is deployed using FastAPI with a static web dashboard, Flutter/Dart mobile client

integration, user authentication, saved reports, smart presets, and dark/light theme support.

The formulas and model descriptions in this paper are aligned with the implemented project. Features such as live weather integration, multilingual advisory, and cloud-based report storage can be added later, but they are not described as completed modules in this paper.

Future Research Directions:

- Integration of live weather APIs or local weather station data.
- IoT-based soil sensor input for automatic NPK and moisture readings.
- Expansion of disease image datasets and full model validation.
- Multilingual advisory generation for regional farmers.
- Cloud storage for reports and multi-user advisory history.

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